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#include <iostream>
#include <cstring>

#define SIZE 100

using namespace std;

class Queue
{
public:
    Queue()
    {
        top = 0;
        bot = 0;
        count = 0;
    }

    int enqueue(int data)
    {
        if (count == SIZE)
        {
            return 0; // Queue overflow (full)
        }
        this->data[bot] = data;
        bot = (bot + 1) % SIZE; // Move to the next position circularly
        count++;
        return 1;
    }

    int *dequeue()
    {
        if (count == 0)
        {
            return NULL; // Queue underflow (empty)
        }
        int *temp = &data[top];
        top = (top + 1) % SIZE; // Move to the next position circularly
        count--;
        return temp;
    }

private:
    int data[SIZE];
    int top, bot, count; // top: front of the queue, bot: rear of the queue, count: number of elements
};

int main()
{
    int data, *temp;
    char command[50];
    Queue *queue = new Queue();

    while (1)
    {
        cin >> command;
        if (strcmp(command, "exit") == 0)
        {
            break;
        }
        else if (strcmp(command, "enqueue") == 0)

```

```

{
    cout << "Please input an integer data: ";
    cin >> data;
    if (queue->enqueue(data) == 1)
    {
        cout << "Successfully enqueued data " << data << " into queue." << endl;
    }
    else
    {
        cout << "Failed to enqueue data into queue (Queue is full)." << endl;
    }
}
else if (strcmp(command, "dequeue") == 0)
{
    temp = queue->dequeue();
    if (temp == NULL)
    {
        cout << "Failed to dequeue a data from queue (Queue is empty)." << endl;
    }
    else
    {
        cout << "Dequeued data " << *temp << " from queue." << endl;
    }
}
}

delete queue;
return 0;
}

```